CHRISTOPHER MORRILL

Norwood, MA | (781) 366-2308 | cmorrill501@gmail.com
linkedin.com/in/christopher-morrill | github.com/cmorrill501

PROFESSIONAL SUMMARY

Detail-oriented and versatile Electrical Engineering graduate with a robust technical background spanning analog circuit design, high-frequency RF systems, custom software simulation development, and machine learning pipelines. Proactive problem solver with proven hands-on capability in PCB prototyping, hardware debugging, and lab equipment verification.

EDUCATION & CERTIFICATIONS

Bachelor of Science in Electrical Engineering | GPA: 3.39/4.0 University of Massachusetts – Boston, MA | September 2022 – May 2026

- **Relevant Coursework:** Electronics I & II, Circuits I & II, Signals & Systems, RF & Microwave Circuits, Antenna Design, Internet of Things, Digital Systems, Fields & Waves

Electronics Technician (Vocational Certificate Program) | GPA: 3.9/4.0 Blue Hills Regional Technical School – Canton, MA | September 2018 – July 2022

- **Core Competencies:** Hardware life cycle execution, digital logic design, circuit troubleshooting, SMT & through-hole precision soldering, manual testing tools.

TECHNICAL SKILLS & QUALIFICATIONS

- **Programming & Data Science:** C++, Python, MATLAB, TensorFlow, Keras, NumPy, Pandas.
- **PCB & Hardware Engineering:** PCB Design, Schematic Capture, PCB Layout, Component-Level Troubleshooting, Hardware Validation, SMT & Through-Hole Soldering, CNC Milling, Chemical Acid-Etching.
- **EDA & Simulation Tools:** KiCad, EasyEDA, TrackMaker, LTSpice, MultiSim, CircuitMaker 2000, AWR Microwave Office, QCustomPlot.
- **Embedded & FPGA Systems:** FPGA Logic Fabric, ARM Architectures, Zybo Z7 SoC, Sensor Integration, Microcontrollers, Synchronous State Machines.
- **Laboratory Equipment:** Oscilloscope, Digital Multimeter (DMM), DC Power Supplies, Spectrum Analyzer, Signal Generator, Logic Analyzer, Solder Rework Station.
- **Core Disciplines:** Circuit Design, Analog Design, Power Electronics, RF/Microwave Design, Signal Processing, Root Cause Analysis, Design Validation, DFM (Design for Manufacturing).

EXPERIENCE

Technical Workshop Instructor – Electronics Fabrication
University of Massachusetts – Boston, MA | April 2026

- Developed laboratory exercises introducing practical PCB assembly and precision electronics manufacturing techniques.
- Designed and prepared an electronics training curriculum for 14 engineering students; successfully trained attending students in fundamental through-hole soldering, safety protocols, and component identification.
- Implemented rigorous visual and electrical quality inspection procedures that resulted in a 100% functional completion rate.
- Maintained laboratory safety compliance, component inventory tracking, and active electronics workstations.

Host
Jake N Joe's – Norwood, MA | October 2022 – April 2024
- Worked effectively in a fast-paced, high-volume operational environment requiring cross-functional teamwork and multi-channel multitasking.
- Maintained strict quality control and order accuracy while coordinating time-critical communication between kitchen staff and front-of-house teams under pressure.

SELECTED TECHNICAL PROJECTS

Custom Analog Circuit Simulation Engine (C++)
- Programmed a 4,000-line object-oriented C++ SPICE simulator engine from scratch capable of solving linear analog circuits and operating point parameters.
- Implemented Modified Nodal Analysis (MNA) and a composite TR-BDF2 integration algorithm to accurately solve time-domain differential equations and model transient RLC circuit behavior.
- Integrated the Eigen linear algebra library to optimize large-scale vector matrix mathematics and execution speed.
- Validated solver accuracy by cross-referencing output node voltages and oscillation peaks against benchmark LTSpice baselines.

Custom Monolithic XL6019 Step-Up Converter PCB
- Designed a 2-layer switching boost converter circuit in KiCad to step up a 3S Li-ion battery supply to a stable 12V output.
- Completed full schematic capture, components sizing selection, and physical layout optimized for a 5A max current draw.
- Routed wide high-current copper pours and optimized the Power Distribution Network (PDN) loop areas to suppress electromagnetic interference (EMI).
- Debugged layout errors during hardware validation by hand-soldering a discrete resistor path, successfully correcting the feedback network loop.
- Tested and verified transient output profiles using a bench oscilloscope, digital multimeter, and variable DC power supplies.

Multimodal Sensor Fusion & Activity Recognition System (AI/ML)
- Built a dual-path Transformer network pipeline in Python utilizing TensorFlow and Keras to classify human activity from time-series sensor data.
- Programmed signal preprocessing modules using NumPy and Pandas to ingest raw CSV telemetry, normalize signal dimensions, and reject environment noise.
- Improved machine learning model performance by conducting a comprehensive model failure analysis and isolating hidden feature sparsity bottlenecks.
- Developed and injected new velocity-vector tracking input features into the decoder path, increasing final model accuracy by over 20%.